Fine-Tuning Large Language Models for Dow 30 Stock-Movement Forecasting

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Abstract

This project fine-tunes three instruction-following language models on the FinGPT Dow 30 forecasting dataset: Qwen2.5-7B-Instruct, Llama-3.1-8B, and DeepSeek-R1-Distill-Llama-8B. Each model was trained with 4-bit QLoRA on a single NVIDIA Tesla T4 and evaluated on all 300 test examples. Qwen achieved the highest binary direction accuracy (52.19%) and the lowest mean inference time (51.77 seconds per sample), DeepSeek achieved the lowest prediction-magnitude mean squared error (8.77), and Llama achieved the strongest mean ROUGE scores. A 24-response human-in-the-loop (HITL) review found consistently correct formatting but also frequent unsupported or incorrectly interpreted financial claims. A controlled Qwen rank experiment showed that increasing LoRA rank from 8 to 16 doubled the trainable parameters without improving direction accuracy or magnitude error. Overall, rank-8 Qwen provided the best practical balance of accuracy, speed, and resource use, although no model produced direction accuracy sufficiently above chance for investment use.

1 Objective

The objective was to adapt modern open-source large language models(LLM) to a structured financial-forecasting task. Given recent company news, price movements, and basic financial ratios, each model had to identify positive developments and potential concerns, predict the following week's stock-price direction and magnitude, and explain its prediction.

Using the Dow 30 dataset, we employed a LoRA-based parameter-efficient fine-tuning approach. We fine-tuned Llama-3.1-8B, DeepSeek-R1-Distill-Llama-8B, and Qwen2.5-7B-Instruct, subsequently conducting quantitative evaluations based on binary classification accuracy, MSE, ROUGE, inference time, and valid sample count, and qualitatively through a Human-in-the-Loop analysis.

2 Hardware and Software Environment

Training and inference ran in Google Colab on a Linux x86_64 runtime with one NVIDIA Tesla T4. Table 1 records the environment captured after the experiments.

Table 1: Training and evaluation environment.

Component	**Value**	**Component**	**Value**
Operating system	Linux 6.6.122+, x86_64	GPU	NVIDIA Tesla T4
GPU memory	14.56 GiB (15,360 MiB)	NVIDIA driver	580.82.07
CUDA	13.0	Python	3.13.15
PyTorch	2.11.0+cu130	Transformers	5.18.0
PEFT	0.21.2	Accelerate	1.15.0
Datasets	5.0.1	Training platform	Google Colab

The base models were loaded in 4-bit form to fit the T4 memory budget. Gradient checkpointing, batch size 1, and gradient accumulation were used to reduce peak memory. The largest measured peak allocated memory was 12.88 GiB for the rank-16 Qwen run.

3 Dow 30 Dataset

The experiment used `FinGPT/fingpt-forecaster-dow30-202305-202405`. The official split was preserved, with no random resplitting. Each row contains a prompt, reference answer, weekly return label, ticker symbol, and forecast period. Prompts combine company descriptions, recent news, price history, and financial ratios. Reference answers use three required sections: positive developments, potential concerns, and prediction with analysis.

Table 2: Dataset characteristics

Split	Rows	Symbols	Weekly periods	Avg. prompt words	Avg. answer words	Parseable refs.
Train	1,230	30	41	819.19	293.86	1,229
Test	300	30	10	836.76	292.77	297

The training period ranges from May 2023 through February 2024, while the test period covers ten weeks from February 25 through May 5, 2024. The test set contains 163 downward and 137 upward labels. Three test reference answers do not satisfy the exact output parser, which limits the maximum paired metric count to 297. Tokenization also differs across model families. After removing sequences longer than 2,048 tokens, the pipeline retained 619 Qwen examples, 803 Llama examples, and 804 DeepSeek examples. This model-dependent filtering is important when comparing training time and loss.

4 Fine-Tuning Method

The baseline configuration used rank 8, alpha 16, dropout 0.1, one epoch, learning rate 5×10^{-5}, batch size 1, gradient accumulation 16, maximum sequence length 2,048, a constant learning-rate schedule, warmup ratio 0.03, `paged_adamw_8bit`, and random seed 42. The effective batch size was 16. A controlled optimization run changed only Qwen's LoRA rank from 8 to 16.

Table 3: Completed training runs

Run	Kept train rows	Rank	Trainable params.	Adapter MiB	Train loss	Runtime (h)	Peak alloc. GiB
Qwen2.5-7B-Instruct	619	8	20,185,088	77.05	1.0865	4.56	12.71
Llama-3.1-8B	803	8	20,971,520	80.06	1.4963	8.23	12.41
DeepSeek-R1-Distill-Llama-8B	804	8	20,971,520	80.06	1.6564	9.62	12.41
Qwen2.5-7B-Instruct (rank 16)	619	16	40,370,176	154.05	1.0864	5.33	12.88

Figure 1 shows logged step loss for the three rank-8 baselines. All runs show an overall decline, but the absolute losses should not be compared as pure model-quality scores because tokenization and retained training examples differ.



Figure 1: Training loss across optimizer steps for the three rank-8 baselines.

5 Evaluation Protocol

All models generated answers for the same 300-example test split with deterministic decoding (`do_sample=false`), at most 512 new tokens, 4-bit model loading, bf16 computation, and seed 42. The parser required all three sections, plus explicit `Prediction:` and `Analysis:` fields in the correct order.

Four automatic measures were calculated on pairs for which both the generated answer and reference were parseable:

- **Binary accuracy**: agreement on upward versus downward movement.
- **MSE**: squared error between signed percentage-range midpoints, such as +1.5 for "up by 1–2%."
- **ROUGE-1, ROUGE-2, and ROUGE-L**: mean F1 overlap over the positive-development, concern, and analysis sections.
- **Inference time**: mean wall-clock generation time per test sample.

The reported `valid_count` is included with every score because malformed generated or reference answers are excluded from paired metrics.

For subjective evaluation, a HITL form sampled six common test cases per model, for 24 reviews across the three baselines and the Qwen rank-16 variant. Each response received six quality scores from 1 to 5, a formatting-compliance decision, and a hallucination flag.

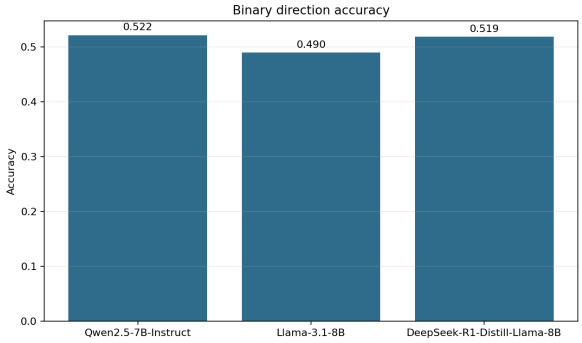
6 Results

6.1 Comparison of the three baseline models

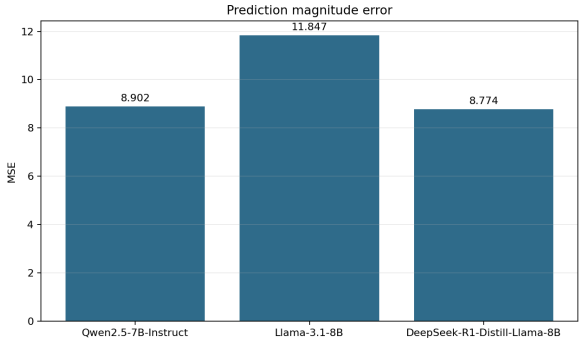
Table 4 summarizes the full-test results. Qwen had the highest direction accuracy and fastest inference, DeepSeek had the lowest MSE, and Llama had the highest mean ROUGE scores. Llama produced one unparsable answer, while Qwen and DeepSeek produced parseable answers for all 300 samples.

Table 4: Baseline evaluation results. Accuracy and text metrics use the stated valid count.

Model	Parseable	Valid	Binary acc.	MSE	ROUGE-1	ROUGE-2	ROUGE-L	Sec./sample
Qwen2.5-7B-Instruct	300/300	297	**0.5219**	8.9024	0.4244	0.1471	0.2527	**51.77**
Llama-3.1-8B	299/300	296	0.4899	11.8471	**0.4367**	**0.1542**	**0.2597**	57.85
DeepSeek-R1-Distill-Llama-8B	300/300	297	0.5185	**8.7744**	0.4283	0.1481	0.2550	61.87



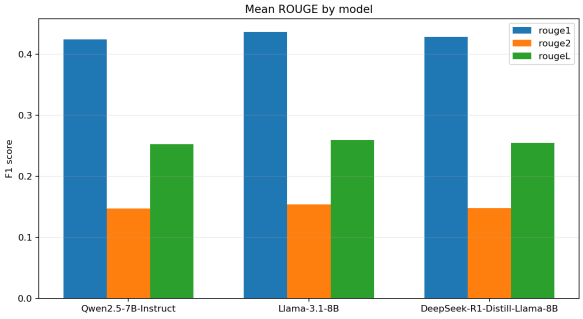
(a) Binary direction accuracy



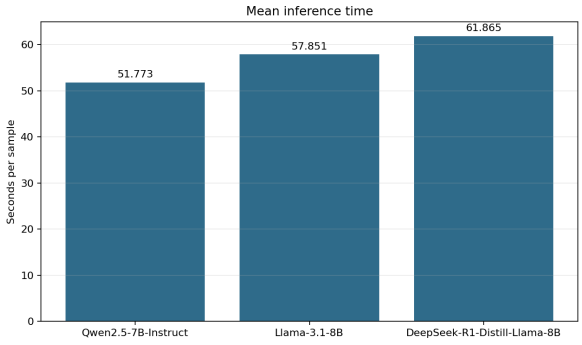
(b) Prediction-magnitude MSE

Figure 2: Prediction results for the three rank-8 baselines.

The accuracy differences are small: Qwen exceeded DeepSeek a little, and all three results are close to 50%. DeepSeek got slightly lower MSE with longer inference time. Llama's stronger ROUGE scores did not translate to magnitude prediction.



(a) Mean ROUGE scores.



(b) Mean inference time.

Figure 3: Text similarity and inference cost for the baseline models.

6.2 Human-in-the-loop review

Table 5 reports the mean HITL scores. All 24 answers followed the required format. Qwen rank 16 received the highest overall quality rating, followed by Qwen rank 8. Qwen also had the lowest hallucination rate, although a 50% flag rate remains substantial. The most common problems were incorrect interpretation of ratios as "high" or "low," unsupported product-market claims, and incorrect attribution of a transaction or investment.

Table 5: HITL results; six reviewed responses per model. Quality columns use a 1–5 scale.

Model	Overall	Positive	Concerns	Financial	Justification	Grounding	Clarity	Format	Hallucination
Qwen2.5-7B-Instruct	3.861	4.333	3.667	3.167	3.667	3.500	4.833	100%	50.0%
Llama-3.1-8B	3.611	3.333	3.667	3.500	3.833	2.833	4.500	100%	66.7%
DeepSeek-R1-Distill-Llama-8B	3.472	3.333	3.500	2.667	4.000	2.667	4.667	100%	66.7%
Qwen2.5-7B-Instruct (rank 16)	**4.028**	4.000	**4.167**	3.333	4.000	**3.667**	**5.000**	100%	50.0%

The HITL results add information that ROUGE and direction accuracy miss. For example, a response may predict the correct direction and follow the template while still misreading a financial ratio. The low factual-grounding means for Llama and DeepSeek are therefore important cautions even though their generated prose is clear and their ROUGE scores are competitive.

6.3 Controlled LoRA-rank optimization

Increasing Qwen's rank from 8 to 16 doubled the trainable parameters and adapter size. Training time increased by 16.9%, from 4.56 to 5.33 hours. Table 6 shows that the larger adapter made small ROUGE gains but slightly reduced binary accuracy and increased MSE. HITL overall quality improved by 0.167, mainly through stronger concern identification and factual grounding, but the hallucination rate did not change.

Table 6: Controlled Qwen rank-8 versus rank-16 comparison.

Run	Trainable params.	Runtime (h)	Accuracy	MSE	R-1	R-2	R-L	Sec./sample	HITL overall
Rank 8	20,185,088	4.56	0.5219	8.9024	0.4244	0.1471	0.2527	51.77	3.861
Rank 16	40,370,176	5.33	0.5185	9.1111	0.4266	0.1481	0.2537	52.23	4.028
Δ (rank 16 − rank 8)	+20,185,088	+0.77	−0.0034	+0.2088	+0.0022	+0.0010	+0.0010	+0.45	+0.167

The rank-16 variant does not provide a convincing automatic-metric improvement proportional to its cost. For deployment on a T4, rank 8 is the more efficient setting.

7 Discussion and Limitations

The experiments identify different strengths rather than one dominant model. Qwen is the strongest operational choice because it combines the best direction accuracy with the fastest inference. DeepSeek has the best magnitude MSE but is 19.5% slower than Qwen. Llama generates language closest to the references but has the weakest numerical forecast metrics. These trade-offs show why a financial-language-model evaluation should include both prediction and generation metrics.

Several limitations constrain the conclusions:

- The test set contains only 300 observations across ten weekly periods.
- Exact-format parsing excludes three malformed references and one Llama generation. Reported scores therefore use 296 or 297 valid pairs rather than all 300 rows.
- Model-specific tokenization retained substantially fewer Qwen training examples, so the training sets are not identical after length filtering.
- ROUGE measures lexical overlap with one reference answer and does not verify financial correctness.
- HITL evaluation covers six cases per model. Its ratings are informative but sensitive to the selected examples and rubric interpretation.
- Inference time is specific to one Tesla T4 and includes the behavior of the tested software stack.
- Direction accuracy near 50% does not support real-money trading or investment decisions.

Future work should test multiple seeds, use a larger time-separated test set, add factual-consistency checks, calibrate magnitude predictions, and compare length-control strategies that retain the same training examples across tokenizers.

8 Conclusion

All three models were successfully fine-tuned and evaluated on the complete Dow 30 test split using a single 16 GB-class GPU. Qwen rank 8 gave the best overall engineering trade-off, DeepSeek gave the lowest prediction-magnitude error, and Llama gave the highest textual-overlap scores. The HITL review showed that clear formatting and fluent prose do not guarantee grounded financial reasoning. Increasing Qwen's LoRA rank to 16 doubled trainable parameters and adapter storage without improving the primary automatic prediction metrics. These results favor the smaller rank-8 adapter and emphasize the need to evaluate factual grounding separately from format compliance and language similarity.